

Global Cybersecurity Threats (2015-2024)

DATA VISUALIZATION TECHNIQUES

Exercise 2

Group 7

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KPI

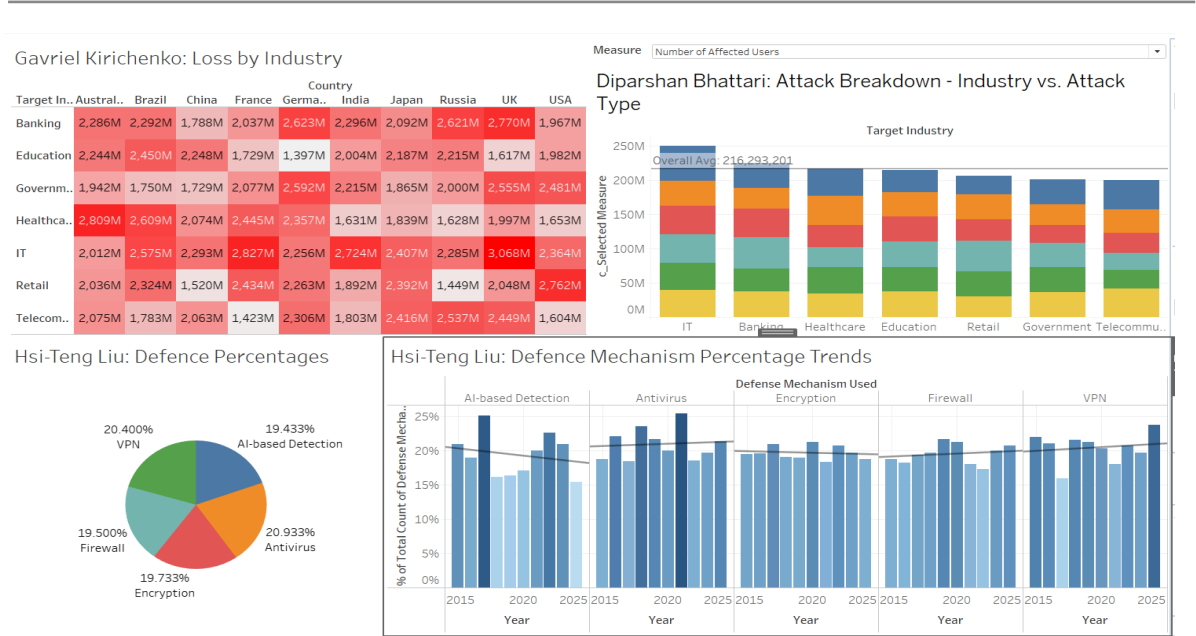
The number of incidents each year.

- The report highlights the total number of cybersecurity incidents recorded globally for each year from 2015 to 2025.

Year-over-year growth percentages of each country.

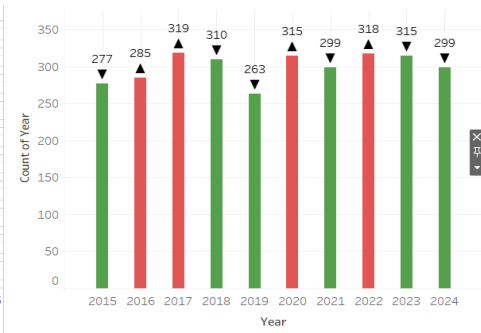
- To gain a deeper understanding of how the cybersecurity landscape evolved in different regions, the report examines the year-over-year growth rate in incidents for each country.

Data Discoveries



This cybersecurity dashboard provides a comprehensive overview of global trends from 2015 to 2025, highlighting financial losses by industry and country, prevalent attack types, and the effectiveness of various defense mechanisms. Healthcare, IT, and Banking sectors face the highest financial impacts, with countries like the UK and India particularly affected. Phishing and ransomware emerge as the most widespread attack types across industries. On the defense side, mechanisms like antivirus, VPN, and AI-based detection are widely used, with antivirus leading slightly. Trends show a gradual increase in the adoption of advanced defense methods over time, indicating improved but still evolving cybersecurity practices.

Country	YOY	YOY	YOY	YOY	YOY	YOY	YOY	YOY	YOY	YOY		
Australia	100.00%	49.23%	-4.19%	11.41%	-26.31%	-15.27%	12.47%	71.81%	-27.28%	2.19%		
Brazil	100.00%	-15.13%	56.28%	-15.24%	-5.64%	-2.05%	41.27%	41.88%	59.70%	7.13%		
China	100.00%	4.49%	-34.94%	13.28%	17.41%	3.00%	21.20%	-9.31%	-10.00%	46.14%		
France	100.00%	-13.74%	46.49%	-16.52%	-1.94%	23.87%	-25.19%	-0.07%	-14.59%	12.62%		
Germany	100.00%	57.82%	-12.90%	-4.53%	-39.24%	37.15%	-9.04%	57.62%	-10.29%	-2.28%		
India	100.00%	-27.34%	23.52%	-10.53%	-7.80%	32.73%	36.73%	40.87%	-0.01%	-26.54%		
Japan	100.00%	-46.63%	147.03%	-17.02%	-33.23%	67.60%	0.92%	3.39%	-6.53%	-27.11%		
Russia	100.00%	-28.79%	39.30%	2.65%	3.53%	-6.93%	18.11%	-30.29%	31.44%			
UK	100.00%	7.52%	-13.56%	-11.87%	12.82%	14.66%	5.75%	1.63%		-20.47%		
USA	100.00%	71.94%	-14.23%	-0.90%	-52.41%	110.71%	-6.47%	11.30%		-12.03%		
		2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025



The chart displays the average cybersecurity ranking for five countries from 2014 to 2025. The Y-axis represents the ranking, ranging from 0 to 40. The X-axis represents the year. Russia's ranking starts at approximately 30 in 2014, dips slightly in 2015, and then shows a sharp increase starting in 2022, reaching approximately 30 by 2024. India's ranking starts at approximately 40 in 2014 and shows a steady decline to approximately 22 by 2024. China's ranking starts at approximately 20 in 2014 and shows a general decline to approximately 18 by 2024. France's ranking starts at approximately 15 in 2014 and shows a steady decline to approximately 8 by 2024. The USA's ranking starts at approximately 10 in 2014 and shows a steady decline to approximately 6 by 2024.

Year	Russia	India	China	France	USA
2014	30	40	20	15	10
2015	31	38	19	14	9
2016	32	36	18	13	8
2017	33	34	17	12	7
2018	34	32	16	11	6
2019	35	30	15	10	5
2020	36	28	14	9	4
2021	37	26	13	8	3
2022	38	24	12	7	2
2023	39	22	11	6	1
2024	30	22	18	8	6

The chart displays the financial loss in millions of dollars for four categories from 2014 to 2025. The Y-axis ranges from 0K to 3K. The X-axis shows the years from 2014 to 2025. The four categories are represented by different colored lines: Cybersecurity (blue), Data Breach (orange), Ransomware (green), and Phishing (red).

Year	Cybersecurity	Data Breach	Ransomware	Phishing
2014	2800	2200	2100	2300
2015	2900	2300	2200	2400
2016	2800	2100	2000	2500
2017	3200	2200	2300	2600
2018	2700	1900	2400	2000
2019	2800	2900	1800	2100
2020	3100	1700	3000	2300
2021	2800	3000	2700	2400
2022	3000	2100	3100	2500
2023	3100	2800	2600	2400
2024	3000	1800	2900	2500
2025	3200	1900	2700	2400

Country	Year	Attack Type						
Australia	2030	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Brazil	2030	Yes	Yes	Yes	Yes	Yes	Yes	Yes
China	2030	Yes	Yes	Yes	Yes	Yes	Yes	Yes
France	2030	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Germany	2030	Yes	Yes	Yes	Yes	Yes	Yes	Yes
India	2030	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Japan	2030	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Russia	2030	Yes	Yes	Yes	Yes	Yes	Yes	Yes
UK	2030	Yes	Yes	Yes	Yes	Yes	Yes	Yes
USA	2030	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Grand Total	2030	Yes	Yes	Yes	Yes	Yes	Yes	Yes

	Security Vulnerability Type / Defense Mechanism Used													
	Zero-day					Unpatched Software			Social Engineering			Weak Passwords		
Avg. Incident Resolution Time (Hours)	36.486													
AI-based Detec.						AI-based Detec.			AI-based Detec.			AI-based Detec.		
Firewall						Firewall			Firewall			Firewall		
Encryption						Encryption			Encryption			Encryption		
VPN						VPN			VPN			VPN		
Antivirus						Antivirus			Antivirus			Antivirus		

Attack Type

Financial Loss (in Million \$)

Target Industry

- Banking
- Education
- Government
- Healthcare
- IT
- Retail
- Telecommunication

Country

- (All)
- Australia
- Brazil
- China
- France
- Germany
- India
- Japan
- United States

Attack Type

- (All)
- DDoS
- Malware
- Man-in-the-Middle
- Phishing
- Ransomware
- SQL Injection

Years: 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022

Attack Type	Target Industry	Country	Year	Cases
DDoS	Banking	(All)	2015	85
			2016	62
		Healthcare	2017	91
			2018	78
			2019	71
			2020	71
	Education	(All)	2021	73
			2022	71
		IT	2015	61
			2016	70
			2017	64
			2018	81
Malware	Banking	(All)	2015	74
			2016	68
		Healthcare	2017	67
			2018	81
			2019	64
			2020	70
	Education	(All)	2021	61
			2022	70
		IT	2015	70
			2016	64
			2017	64
			2018	81
Man-in-the-Middle	Banking	(All)	2015	56
			2016	70
		Healthcare	2017	80
			2018	58
			2019	53
			2020	65
	Education	(All)	2021	77
			2022	77
		IT	2015	77
			2016	65
			2017	65
			2018	58
Phishing	Banking	(All)	2015	51
			2016	89
		Healthcare	2017	89
			2018	63
			2019	68
			2020	73
	Education	(All)	2021	96
			2022	96
		IT	2015	96
			2016	73
			2017	73
			2018	68
Ransomware	Banking	(All)	2015	59
			2016	71
		Healthcare	2017	74
			2018	77
			2019	72
			2020	71
	Education	(All)	2021	69
			2022	69
		IT	2015	71
			2016	71
			2017	71
			2018	71
SQL Injection	Banking	(All)	2015	78
			2016	63
		Healthcare	2017	77
			2018	72
			2019	75
			2020	67
	Education	(All)	2021	71
			2022	71
		IT	2015	71
			2016	71
			2017	71
			2018	71

This dashboard presents a detailed analysis of global cyberattack patterns, financial impacts, and defense effectiveness. It shows the distribution of attack types across countries, with DDoS, Phishing, and Ransomware being the most prevalent. The economic impact chart highlights that Phishing and Ransomware attacks are the most financially damaging, each accounting for the highest number of cases and losses exceeding \$ 27 M. In terms of defense, the analysis of vulnerabilities such as zero-day exploits and unpatched software reveals that firewalls and antivirus software tend to resolve incidents most effectively. The visualizations collectively emphasize the need for proactive defense strategies tailored to evolving threats and vulnerabilities.